RAKESH ALGOT

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BACKEND ENGINEER | AI SYSTEMS & LLM INFRASTRUCTURE

Backend Engineer with experience building scalable APIs and asynchronous backend systems using Python and FastAPI. Worked on AI/LLM integrations including RAG pipelines and retrieval optimization, with a focus on performance and reliability in production environments.

CORE SKILLS

Backend & API Development: Python, FastAPI, REST APIs, API Design, Async Processing.

Databases & Caching: MongoDB, Redis, Query Optimization, Indexing, Caching Strategies.

Systems & Infrastructure: Docker, Linux, Distributed Systems, High-Throughput Processing.

AI Systems & Retrieval: RAG Pipelines, Hybrid Retrieval (Dense + BM25), Reranking, Chunking.

LLM Systems: Prompt Engineering, Structured Outputs, Evaluation, Validation Pipelines.

Vector Search & Data Systems: Qdrant, FAISS.

PROFESSIONAL EXPERIENCE

Backend Engineer

Yensi Solutions — Hyderabad | 2024 – Present

- Contributed to the design and development of backend pipelines for data retrieval and processing in production systems
- Developed REST APIs using FastAPI with structured logging and role-based access control, and Docker-based deployment
- Reduced API response latency by ~40% through Redis caching and query optimization
- Designed and implemented asynchronous worker pipelines for high-volume data ingestion and background processing across large document datasets
- Improved retrieval performance and response quality by optimizing ranking pipelines (~20–30% improvement)
- Reduced system costs by optimizing local processing and minimizing external API dependencies
- Integrated retrieval-based AI/LLM components into backend services for intelligent data processing

PROJECTS

MedBridge — AI-Powered Medical Record Intelligence Platform

Tech: FastAPI, MongoDB, Docker, AsyncIO, OCR/LLM Pipelines

- Developed backend services for medical record digitization and structured data management
- Worked on asynchronous AI/OCR pipelines for document ingestion, focusing on job orchestration and processing workflows
- Developed secure APIs for document access, doctor sharing, and multilingual patient queries

Citex — Citation-grade Context Retrieval System

Tech: FastAPI, Qdrant, Docker, Sentence Transformers, FastEmbed (BM25), FlashRank, LangChain

- Worked on hybrid retrieval pipeline combining dense vector search and BM25 with reranking for improved result relevance
- Contributed to ingestion and indexing pipelines, including chunking, metadata handling, and deduplication
- Implemented Redis caching to optimize repeated queries across large document datasets

ClubOS — After-School Club Operations Platform

Tech: FastAPI, MongoDB, Redis, Docker, Stripe

- Designed and implemented backend APIs for bookings, attendance, and payments within a multi-tenant system
- Implemented asynchronous processing for background tasks such as bookings and payment handling
- Applied Redis caching to optimize frequently accessed data and improve response times
- Improved system reliability and performance for multi-site operations through optimized query handling and backend workflows

EDUCATION

Master of Computer Applications (MCA)

Nizam College — 2022